6

Workstations

So far in this text we have discussed some of the overarching design approaches and traditional automation formats. We have looked at cost justification methods, and the impact of estimating costs. However, we have not looked at any real level of detail in the design of our automation. Because every automatic machine can be different from the next one, one will need to look at the discussions in this chapter as a series of examples, while trying not to force your current viable concept into a rigid mold.

We will use the term "Workstations" as a catch all to some of these missing details. A workstation is where the automation does something, whether by a dedicated machine, an industrial robot, or even a human operator if it is a process step that cannot be successfully handled any other way. Workstations are a logical grouping of commercially available components, integrated with a series of custom design parts and/or electronics. They is more than a single motor or an air cylinder. Workstations are usually connected either by conveyors, a rotary index system, or any of the other options discussed in Chapters 4 and 5.

To best understand the specifics of a workstation, one could argue that the topics in later chapters on actuators, sensors, and controllers need to be covered first. In general, the order of the topics in this and the next four chapters are similar to the old statement, "which came first, the chicken or the egg." What one finds in designing automation is that the process is not linear. Solution concepts sometimes come coupled to a new actuator, or an implied control strategy. Students using this text may find themselves iterating through these topics several

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times until it all synchronizes together. With this said, there will be many small points on actuators and sensors that will be mentioned in this chapter that will hopefully make more sense when their details are covered later.

It should be noted that not everyone in the automation field uses the term "workstation." If one searches the Web for workstations, one will either find an efficiently laid out workbench for a human to process some manufacturing task, or they will find some laboratory automation workcells used in the pharmaceutical market for drug discovery. One needs to search the Web for the specific function required at that workstation. However, it is logical to approach the next phase of automation design by grouping technology using this somewhat arbitrary naming convention.

There are many commercial workstations available. Many can be found by determining the specific function one wants to incorporate into a machine. Samples of these include:

- air-powered press;
- screwdriver;
- hot glue melt;
- automatic fill heads for liquids;
- ultrasonic welding.

Rarely does one want to start the design process by replicating these. Because there is usually more than one vendor able to supply these often-used devices, competition normally keeps the pricing reasonable. It is foolish to think that one could design a hot glue melt applicator cheaper than the systems that have been around for 30 years or more. However, that never stops some engineers, who want to design the entire machine, even at the risk of higher costs and greater delays!

A custom workstation would be the actual detailed design of the device to tie bows on packages, as discussed in Chapter 2. During the required actions described in Chapter 2, a concept on how to do the process was established, but none of the real details as fleshed out. A gripper was seen as a desired tool in the process steps, but we did not determine if the gripper was to be placed on a robot or a custom designed pick and place device. Also, the gripping surfaces need to firmly hold the ribbon without destroying its polished appearance.

The determination of whether or not to use an available workstation will often take the following steps:

1. Look for available devices.
2. Determine all of the combinations of workcell layouts and the needed workstations.
3. Play "what if" games if one is not limited to available devices and can make custom devices.

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4. Establish a trade-off matrix to compare all options.
5. Use the "best" answer, but always remember the other options. The best answer may be controlled by nontechnical issues such as economic conditions, safety, union contracts, and so on.

It could be that by using commercially available workstations, a designer is painted into a corner to create a solution similar to every other company's attempt at automation. A novel concept may drive the overall system layout to a better solution.

In the case of tying a bow, if one could find such a workstation, the task would be moot. It is the key problem to this project. As for determining all possible combinations, a quick hand sketch of what the different designs would look like would be very useful. As one might imagine, the work discussed in Chapters 2, 4, 5, and this chapter are almost always an iterative process. If after several hours or days of automation concept development, no possible gripper design to grab ribbon and pull it through the loop can be found commercially or custom designed, then perhaps the great idea in Chapter 2 is just a dream. One must bring all of these steps into focus, or else one needs to look at the second best design, or maybe even will have to say that it cannot be done!

So since this is an iterative process dealing with multiple chapters of this book, not all automation engineers will proceed with such linear thinking. They may jump to an answer based on earlier successes with other applications, or they may be well versed in one particular robot language, and see that solution path as being the simplest and less risky path to take. Is it the best path overall? Sometimes the answer is "yes" and sometimes the answer is "no".

6.1. WHEN IS IT A WORKSTATION?

So with the above introduction, we could use these checkpoints to assist with defining the tasks of a workstation:

- something that does some operation;
- does it repetitively;
- uses some type of power.

Other issues for workstation designs to consider are:

- Do you perform the operation online or remove it from the conveyor?
- Do you consider a few complex motions or many simpler motions?
- Do you integrate structural members with the actuators?
- What are the roles of sensors?
- What kind of accuracy is needed?

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- What kind of adjustments will be needed, and how will they be done?
- What kind of safety standards will you have to adhere to?

With these definitions and raised concerns, one would likely assume that a conveyor to move a stream of products from point A to point B would not be a workstation. And in most cases this is true. But every now and again, some external device will be integrated with the same type of conveyor, and just like the confusion over what is and is not a robot, a workstation is born. Again, if the customer wants to call a conveyor system a workstation, and they are happy with your price, why not go along?

6.2. WORKSTATION BASICS

The building blocks of most workstations will include:

- structural members;
- bearing devices;
- drive mechanisms;
- actuators;
- sensors.

One might ask about the distinction between a drive mechanism and an actuator. A drive mechanism could be a gearbox or chain and sprocket system that takes the motion and power generated by an actuator (an electric motor for example) and produces the appropriate torques and speeds. Actuators will be covered in Chapter 8, so they will be generally left out of the discussion here. Likewise, sensors will be discussed in Chapter 9.

Many automation builders do not always divide the total machine into workstations; they just build the machine in segments of the overall machine, but the trade-offs and issues of this chapter still remain. A designer needs to know what components can be purchased and what needs to be built from scratch, and how to effectively integrate and interface them. And the laws of physics cannot be ignored.

6.2.1. Structural Members

If you work alongside automation builders long enough, someone is bound to make the statement, "well, it doesn't have to fly." This implies that the strict weight restriction that aircraft designers need to adhere to is not a concern in the automation world. Since the machine is probably shipped by truck only once in its life, the total weight is usually allowed to accumulate faster than one could ever believe, particularly the machine frame. No-one wants the frame to bend or buckle. And since the frame is often designed before the entire machine is laid out, one is not sure what the static or dynamic loads will be (Norton, 1999). This is not the best practice, however.

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The Frame

Many machine frames are constructed from steel — lots of steel. If it is to be located in a food processing plant, it will need to get washed down once each day and therefore is likely to be stainless steel. Rectangular hollow steel tubing, perhaps 2 in. × 2 in., that could be welded into something that might support a military tank is a natural reaction from experienced machine builders.

Sometimes the thought of bolting steel members together sounds a little more flexible in the design phase when there is still some doubt of exactly where some facet of the machine needs to be located, but since most automation machines will be humming along for years to come, the thought of bolts coming loose from the vibrations gives experienced designers concern.

Another concern for the frame (as well as the entire machine) would be the effects of vibration and any resonance modes. These resonance modes can be one of the biggest surprises after a piece of automation has been built and is undergoing test runs. Sometimes the resonance occurs at the normal operating speed, but more likely it can happen when the machine is accelerated from rest up to the operating speed. If the resonance transition time is short, and there is little that can be done to stiffen the frame and moving members, it may be deemed to be acceptable. Or it may shake the entire machine apart in six months. This is something that some computer analysis can assist with, but the level of modeling required still may produce simulation results that are shifted significantly from the real-life occurrences.

The support frame could be extruded aluminum components, since they are available from more than a dozen suppliers. They are easy to order, and some brands have a fantastic number of stock items for joining lengths of aluminum extrusions to form a structure or to have some limited motion (perhaps an access door). Various clear plastic sheets can be cut to fit within the frame, giving a very professional look in a very short amount of time. However, these members, being aluminum, are not as strong or rigid as steel. And they are not welded together, they are bolted together. They will deflect more than one might want, and are often only used for external guarding or very limited load-bearing situations.

Whatever materials and construction are used, the frame will be custom designed and built for your machine. One cannot purchase a meaningful frame from a catalog.

Moving Members

The mass of the moving members does have an impact on motor selection, overall speed, and the response of the machine. So a greater effort is taken to try to minimize the weight and inertia effects. Most automation builders will use a very detailed CAD drawing to determine the size and assembly characteristics of the moving members. They will use the CAD program to calculate weight

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and inertia properties for them. Only some automation builders will actually run a finite element analysis (FEA) on the moving members to see what kind of deflections will occur. For some machines, deflections may not be critical due to relatively slow speeds, but for other machines, one should do the FEA so as to limit any surprises. Not enough machine design houses feel the need or are comfortable performing FEA.

Most moving members are custom designed and built, either in-house or at a subcontractor to a machining builder.

6.2.2. Bearing Devices

The frame and moving members are usually obvious to even the novice automation designer, but when it comes to attaching the moving members to the frame, some people might not think enough about friction, wear, and the need for bearings. And one cannot just look at standard roller bearings similar to what is found in one's automobile wheels. It all depends on the speeds and loading situation. Something will become a consumable, either lubrication that will be added periodically, or parts that will need to be replaced on a specified timetable.

Much of the concern is based on friction between two surfaces. Figure 6.1 shows two scenarios, where the magnified large peaks on the left will potentially have less frictional effect than the lesser peaks on the right due to the reduced contact area. This is why engineers sometimes guess incorrectly on what will improved a high frictional situation. However, when it comes to bearings and bushings, the profiles will hopefully look significantly smoother than either scenario.

Loads are generically found being applied to rotating shafts. This is because there are often a great number of rotating shafts involved with the major operations of the automation. The loading can be radial (Fig. 6.2), thrust (Fig. 6.3), or a combination (Fig. 6.4). The types of bearings will differ signifi-

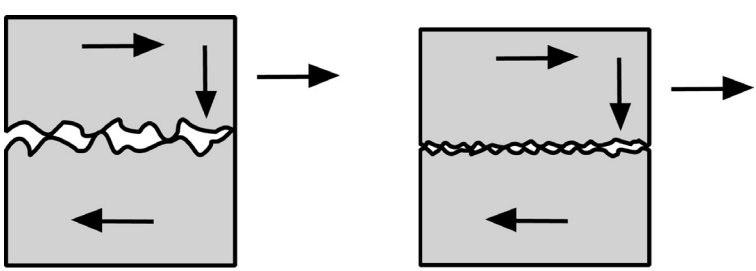


FIG. 6.1 Magnified frictional surfaces; (a) very rough; (b) moderately rough

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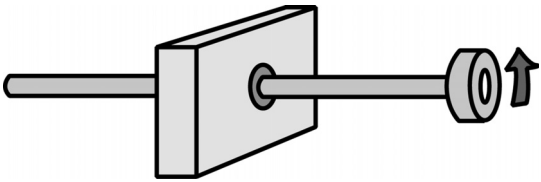


FIG. 6.2 Radial loading of rotating shaft

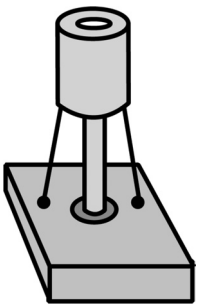


FIG. 6.3 Thrust loading of rotating shaft

cantly for these three cases, so one cannot simply guess and select a bearing that happens to be sitting on the shelf nearby.

The form of the bearing can take several forms for each of these loading situations: thrust bearings (Fig. 6.5) with washers (Fig. 6.6); sleeve bearings (Fig. 6.7); and flanged bearings (Fig. 6.8). And then there is the more well-known roller bearings (Fig. 6.9), with their several internal components (Fig. 6.10).

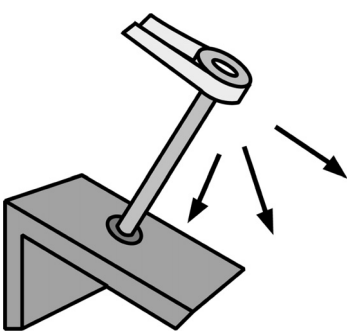


FIG. 6.4 Combined loading of rotating shaft

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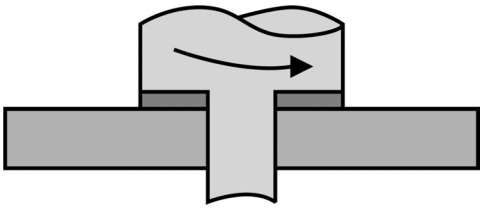


FIG. 6.5 Thrust bearing

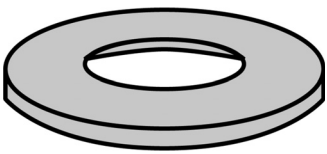


FIG. 6.6 Bearing washer

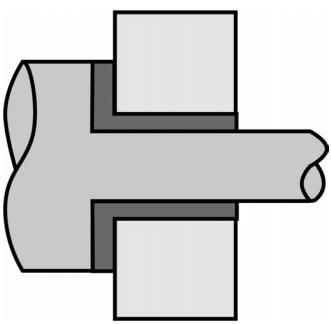


FIG. 6.7 Sleeve bearing

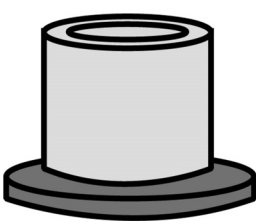


FIG. 6.8 Flanged bearing

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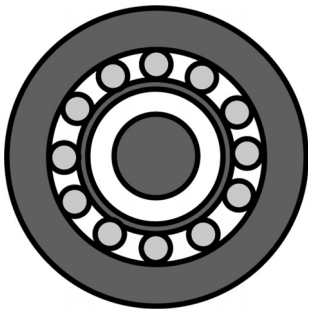


FIG. 6.9 Roller bearing

If the size of the bearing needs to be very compact, sometimes designers will opt for needle bearings. Figure 6.11 shows a cutaway view of such a bearing. If the loading is of greater forces, or there is combined loading, then the options are cylindrical (Fig. 6.12), tapered (Fig. 6.13), or spherical (Fig. 6.14). Spherical offers the ability for the shaft and bearing to be self-aligning, something that may be well worth the extra costs. This allows for accumulated errors in machining and assembly to be accommodated when the automation machine might be 95% completed.

Sometimes an older more traditional bearing surface will work just as well. A sleeve bushing, made of material that will either create a hydraulic film of lubrication, or will run in a dry condition, will be useful. Figure 6.15 shows a standard sleeve bearing. These can be press fit into the frame or other moving member, and their compact attachment properties can be attractive (Fig. 6.16).

Other times, sleeve bearings are placed into pillow blocks (Fig. 6.17). Another option is to replace the pillow blocks with a split sleeve bearing made in

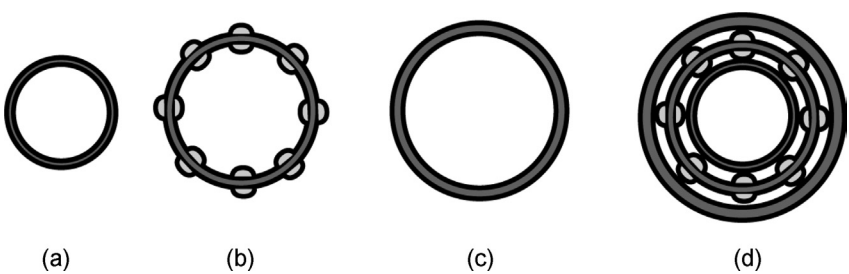


FIG. 6.10 Roller bearing: (a) inner ring; (b) race with embedded roller balls; (c) outer ring; (d) assembled unit

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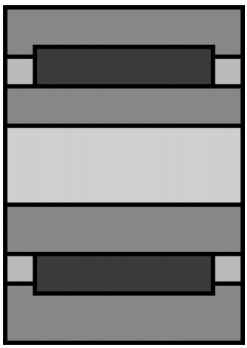


FIG. 6.11 Needle bearing: cutaway view

two halves (Fig. 6.18) to assist with assembly. No one wants to force a six-foot long shaft through a tight sleeve bearing, or worse yet, several tight sleeve bearings.

If the shaft or rod is to reciprocate in the bearing block, then there is a linear bearing available that is commonly used. There are several versions out there, but the classic version uses ball bearings that recirculate through channels (Fig. 6.19) to allow proper load reactions between the balls and the reciprocating motion of the rod, and the bearing block. These bearing are more expensive than roller ball bearings. But when the application calls for them, they are the right solution for the task.

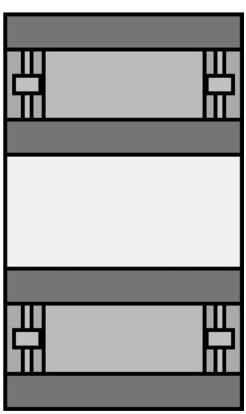


FIG. 6.12 Cylindrical bearing: cutaway view (cylinders with end pins spin within race)

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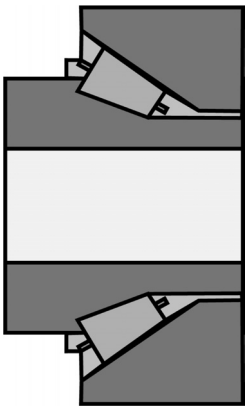


FIG. 6.13 Tapered roller bearing: cutaway view

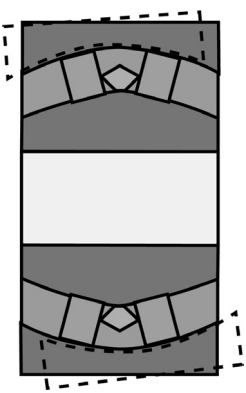


FIG. 6.14 Spherical bearing: cutaway view (dashed lines show alignment possibilities)

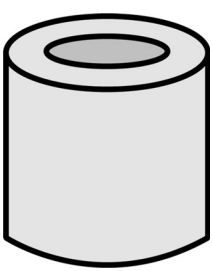


FIG. 6.15 Sleeve bearing

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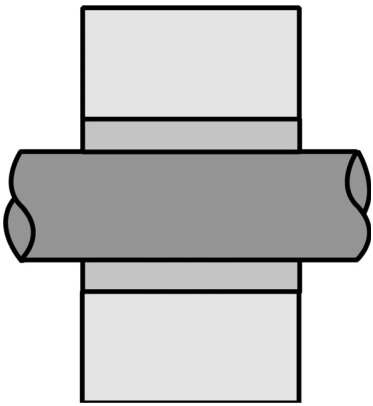


FIG. 6.16 Sleeve bearing press fit into support frame

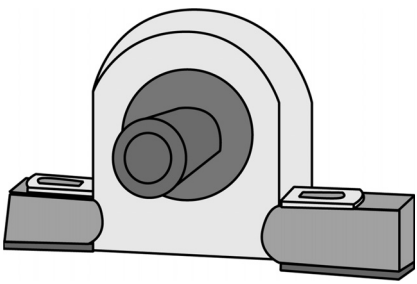


FIG. 6.17 Pillow block

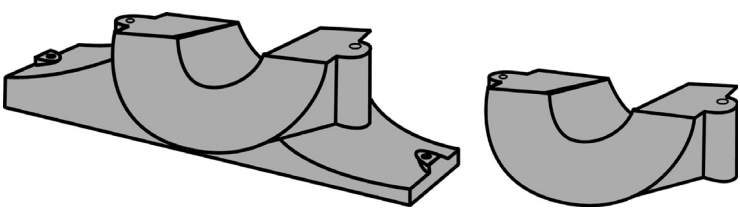


FIG. 6.18 Split sleeve bearing

6.3. DRIVE MECHANISMS

The drive mechanism takes the actuator output motion and transforms it to another location, power ratio, or manipulates it to some other characteristic. It could be a standard drive belt and pulley system (Fig. 6.20), either with similar sized pulleys or different sizes to change the speed and torque ratio. Belt sizes and types are stan-

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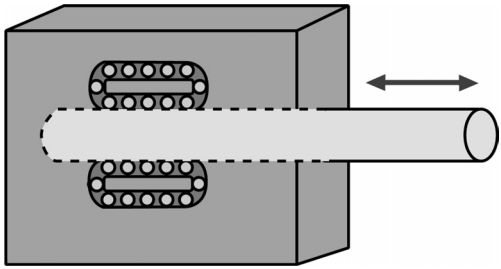


FIG. 6.19 Linear bearing



FIG. 6.20 Drive belt and pulley system

dardized, with the most common being the V belt (Fig. 6.21). There can be an idler pulley to take up the slack and reduce wear and noise (Fig. 6.22).

There are chain drives (Fig. 6.23) with which most people with two wheel bicycles can claim familiarity. As with a bike, most chains need lubrication. So their use is often coupled with a maintenance schedule to check on lubrication, and the need to clean up excess grease, but chains do have the added benefit over belts in that there is no slipping between the two shafts. The chain can stretch and wear out, but the relative timing can be assured.

A more up-to-date version of a chain, which is much quieter and can have fewer maintenance issues, is a timing belt (Fig. 6.24). The belt has teeth molded

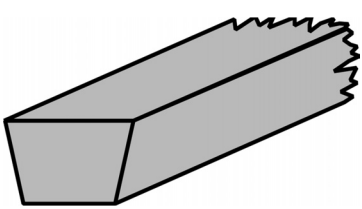


FIG. 6.21 Standard V belt

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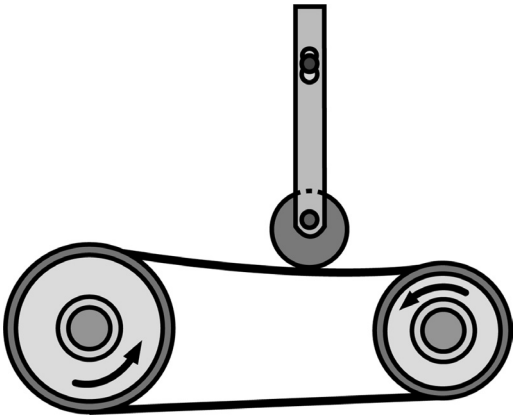


FIG. 6.22 Belt drive with idler pulley

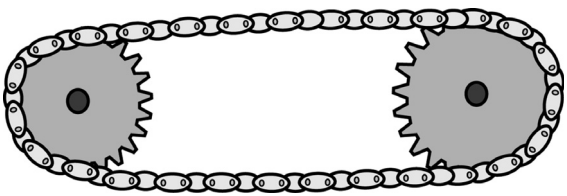


FIG. 6.23 Chain drive

into the inside surface, and the pulleys have matching surfaces that keep the belt registered. So as long as the tension is maintained (and timing belts can stretch from use) the belt should not slip. The allowable amount of force transmittable is given by each supplier.

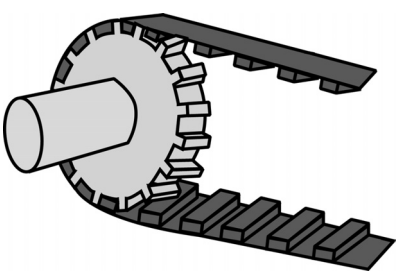


FIG. 6.24 Timing belt

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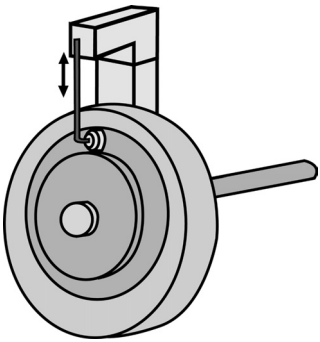


FIG. 6.25 Cam system

A traditional method used by automation designers of old is the cam (Fig. 6.25). It is still very valid today, but it lacks the flexibility or reprogrammability that is available through today's controllers and actuators. There is a significant cost in milling out the cam groove accurately, and the groove surface is costly to repair if it gets worn or damaged. However, the resulting motion can be very deterministic for many operational cycles.

Other drive mechanism issues consist of brakes, such as a band brake (Fig. 6.26) and a combined clutch and brake (Fig. 6.27). These items are needed when the machine's large rotation inertia needs to come to rest quicker than just letting it slow down on its own. Safety reasons alone may require a braking system. If it uses the right kind of bearings, it may take quite a while to stop without some assistance.

The last area to consider is the joining of two shafts. If one does not think too hard about this problem, they are likely to use a simple rigid shaft coupling as in

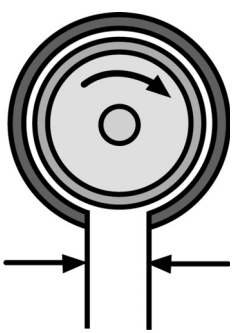


FIG. 6.26 Brake band to stop rotating shaft

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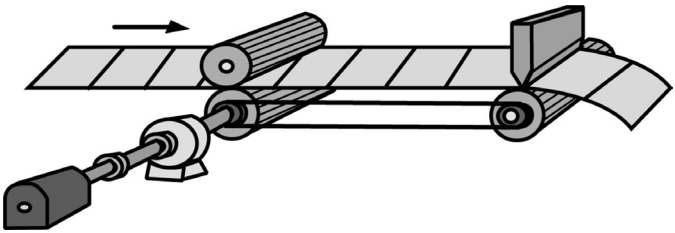


FIG. 6.27 Clutch brake system

Fig. 6.28. This is fine if the shafts are to rotate once per hour, or something similarly eternally slow, but for any rotational motion greater than this, the stress on each of the two shafts, and their bearings and support frame, will lead to trouble.

It is better to use either a universal joint (Fig. 6.29), when one knows that there will be significant shaft misalignment, or some other type of helical cut shaft coupling (Fig. 6.30) if the misalignment is much less. The helical joints can torsionally wind up when significant loads are applied to a coupling that is perhaps under-rated, but the benefits outweigh the new problems created.

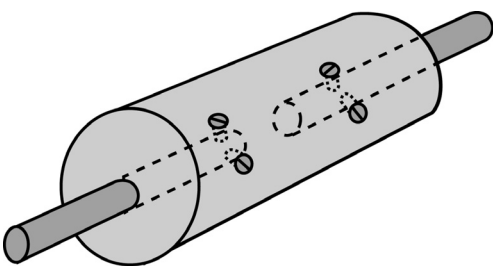


FIG. 6.28 Rigid shaft coupling

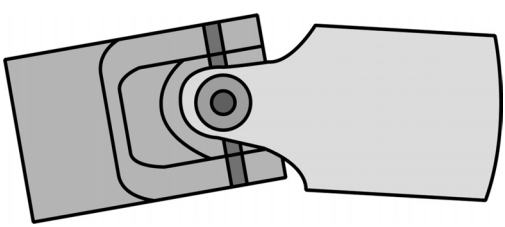


FIG. 6.29 Universal joint

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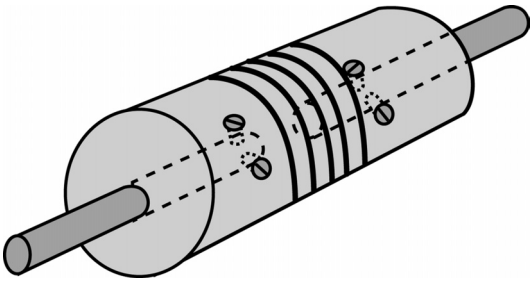


FIG. 6.30 Helical coupling joint

The misalignment of two shafts with high stiffness bearings can result in huge loads, which equal a loss of energy and a reduction of life.

6.4. CASE STUDY NUMBER 1: TBBL WORKSTATION DESIGN

If we now consider some of the details of the books on tape case study in Sec. 4.7.4, there is a list of questions to be first developed and then hopefully answered. Figure 4.23, repeated here (Fig. 6.31), shows some required motions to align either of the two different thicknesses of cases and then firmly hold the case in place for the latch unfastening action to occur. This was determined to be easier than adjusting the latch unfastening device vertically to accommodate the two different thicknesses.

So how is the case lifted up? And when does this happen? The case is probably moving forward by a conveyor system, as per the linear asynchronous design in Chapter 4. A hard stop, either fixed or sliding into the path of the tape case, will align the case by its leading edge (Fig. 6.32). Then the next step can occur if the automation system knows the hard stop was actually hit. A light beam (Fig. 6.33)

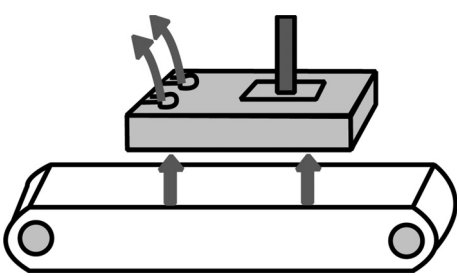


FIG. 6.31 Motions to lift and unlatch a case

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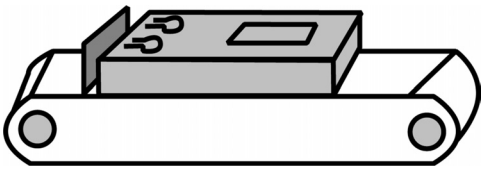


FIG. 6.32 Movable hard stop aligning leading edge of a case

or limit switch (Fig. 6.34) can be used as sensors to tell the controller this has happened, among other options. These two options are most likely to be cost effective compared to say a fully outfitted computer vision system, which would work fine but likely be overkill. Each of these sensors has its pluses and minuses (left for discussion in Chapter 9 on sensors), but they are fairly interchangeable for our planning here.

So how can the case now be lifted up? The motion would logically seem to be pure translation, so a series of rotations would be possible but lacking in efficiency. We could lift by:

- moving the case by vertical translation or sliding;
- using a scissors jack approach;
- inflating an air bladder;
- using magnetic power;
- using a four-bar linkage.

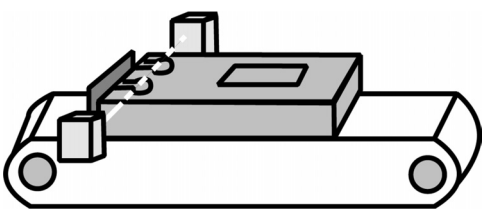


FIG. 6.33 Use of an electric eye beam to detect a case

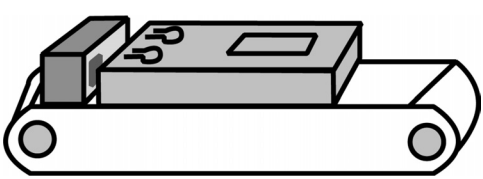


FIG. 6.34 Limit switch on hard stop to detect case presence

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For some of these options, the form of an actuator (or power) is stated, as with the air bladder, but for others, like the scissors jack, the discussion on types of actuators needs to continue.

6.4.1. Moving the Case by Translation or Sliding

Because the lifting motion is not very high, and the case is fairly light, perhaps the easiest thing to do is to eliminate some possibilities first. We can therefore rule out the need for hydraulics, using its pressurized oil like an excavating backhoe, since this type of actuator would be used for lifting tons of mass. And since the lifting needs to occur only until the top edge has reached a target height, we do not need to consider a servo feedback driven system, which is more complex and would cost more.

We can consider compressed air, as both a motion strategy and a power source. The compressed air (or pneumatic) cylinder can lift (Fig. 6.35) until the reaction force of the lifting against a hard stop balances the internal air pressure (Fig. 6.36). It should not damage the case unless the supplied air power is excessive. It could easily handle an infinite number of case thicknesses without any significant sensory feedback or controller requirements. If compressed air is available, it is a potentially ideal solution.

A side issue would be the structural requirements for the lifting. Many air cylinders are designed to only handle loads along the axis of motion. Any side loading will quickly wear out the cylinder. And the concept in Figs 6.35 and 6.36 are idealized, assuming that the center of mass of the load lifted is directly about the cylinder. However, there may be an odd number of tapes in the case, so this assumption may not be true.

It would be better to have a pair of linear bearing rods and supports integrated into the design. Figure 6.37 shows one possibility. This looks easy enough, but for those who have tried it, getting the air cylinder and the two linear rods aligned for relatively frictionless motion through the entire range of motion is

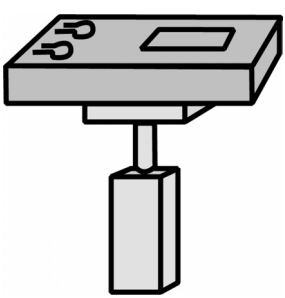


FIG. 6.35 Air cylinder for lifting

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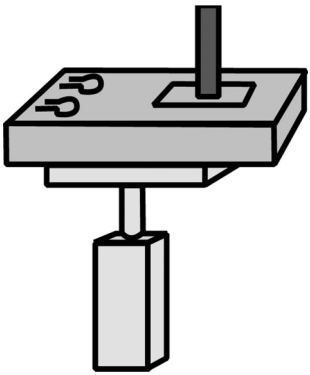


FIG. 6.36 Lifting until reaching a hard stop

not trivial. So, many engineers select a linear slide that is an air cylinder integrated with the two linear bearing rods as a single component (Fig. 6.38). A plate to distribute the support under all of the case would be attached. It is then almost impossible to apply the component incorrectly (although the author has seen it done).

A second option to the air cylinder would be a screw drive powered by a DC electric motor. This motor need not have any complex feedback associated with it. There would need to be a limit switch or electric eye beam similar to the case registering to the hard stop in Figs. 6.3 and 6.4. For the screw drive, there is a need for at least a single load-bearing rod, so as to restrict the case lifter to not spin (Fig. 6.39). A second rod is usually used for balancing forces (Fig. 6.40). Most screw drives units are not designed to see significant side loading or off-axis torques, since these will create early product failures.

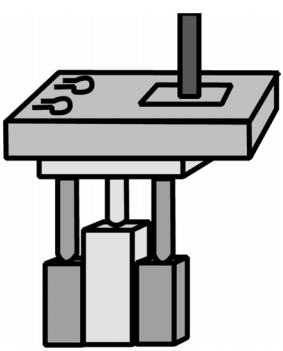


FIG. 6.37 Two load-bearing rods located in parallel to the air cylinder

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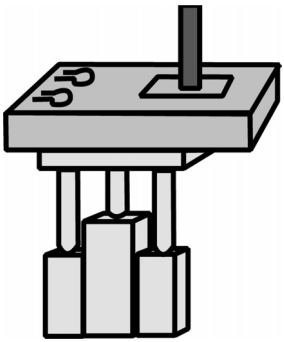


FIG. 6.38 Integrated air cylinder and load-bearing rods

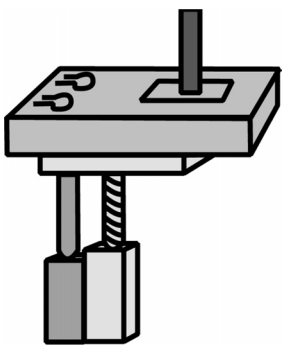


FIG. 6.39 Screw drive and the resistant to twist single bearing rod

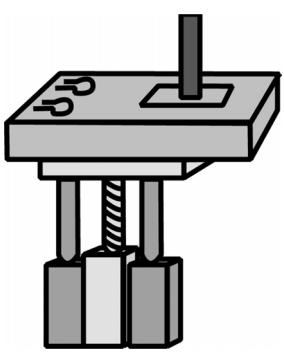


FIG. 6.40 Screw drive with a pair of bearing rods

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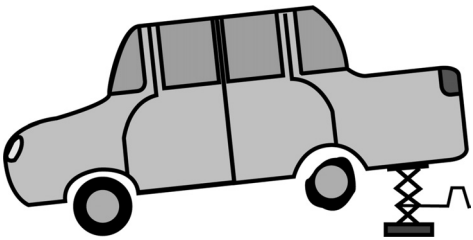


FIG. 6.41 Scissors jack lifting a car for tire change

This option will probably cost a few more dollars than the air cylinder approach, but the cost differential is not much. If there is no compressed air available, or if compressed air causes a problem for other reasons, the electric motor screw drive could be the best choice.

6.4.2. Use a Scissors Jack Approach

Some cars use a scissors type jack to lift a car when changing a tire. Figure 6.41 shows one in use. The horizontal threaded screw shaft and nut are designed such that the jack is self-locking; that is, the frictional forces do not allow the weight of the car to back drive the screw and handle to spin freely. This is almost always a good thing when applying a scissor jack.

Using a scissors jack to lift the case in our workstation (Fig. 6.42) has some benefits. The top of the jack is held quite parallel to the reference surface of the jack's bottom frame member. The case would be effectively supported under the latch unfastening device. The actuator drive would most likely be a DC motor to spin the screw shaft. There are rotational air-powered motors, but they are less commonly found, and can be noisy.

The scissors jack has more moving parts, and might be more expensive that the linear slide units, but the scissors jack might be more compact vertically than

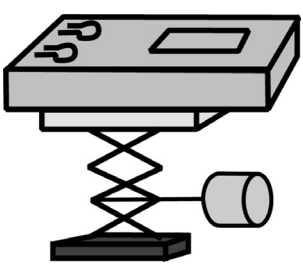


FIG. 6.42 Scissors jack lifting the case

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the linear slides. This dimension may drive the automation's base frame to a lower reference surface, assuming that the vertical height of the incoming case has been set as a requirement (and most likely would be). Yes, a conveyor system could be used to lift the cases up so as to be over a taller lifting device, but this adds to the cost that is possibly not needed.

One will not know which type of lifting device to implement until one finds candidate commercially available units and investigates their dimensions, but there are still other options to explore.

6.4.3. Inflate an Air Bladder

Many trucks on the road today use a series of air-filled bladders to act as both a leveling device and a shock absorber. The lifting function is not as precise as a feedback-driven actuator, but does compensate for uneven loading within the truck's body. These bladders can be a bulging cylindrical or a bellows design (Fig. 6.43). Some packaging machines (Davis, 1997) use similar bladders to cushion the multiple products being manipulated. Imagine the need to absorb the shock when a glass container filled with liquid is inserted into a box by controlling its dropping motion only in the slightest way (Fig. 6.44).

To use an air bladder of either design to lift the books on tape case would require some additional moving structure. The relative motion of the top surface of the bladder, and therefore the case itself, is not well defined (Fig. 6.45). The case most likely would not be aligned properly under the latch unfastening device. Several linear bearing rods would be a viable solution, but then the function of the air bladder has been reduced to that of an air cylinder, which will in general be a fraction of the costs of the bladder. And the air bladder's rubber makeup is not as industrially hardened as an air cylinder. So we will not select this option.

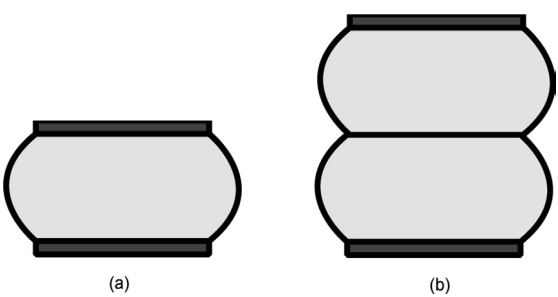


FIG. 6.43 Air bladders: (a) bulging cylindrical; (b) bellows design

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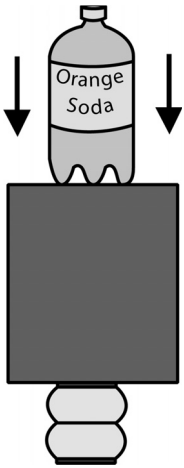


FIG. 6.44 Air bladder cushioning bottle being dropped into case

6.4.4. Using Magnetic Power

One has several choices when it comes to magnetic actuation, but they are in real life limited to several specific niche operations. The main choices are:

- solenoids;
- magnetic levitation.

Solenoids are used in many products found in industry and around the home. The door that holds in the soap until the correct timing of the cleaning cycle of your dishwasher is actuated by a solenoid. They are cheap, small, and can work a reasonable number of cycles. They are made from a wound coil of wire that creates an electromagnet. The center core is a material that responds

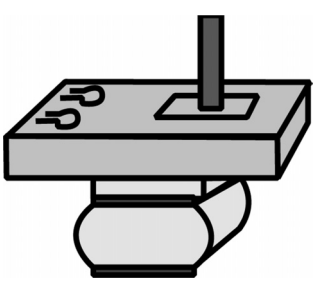


FIG. 6.45 Air bladder lifting a case: not well positioned horizontally

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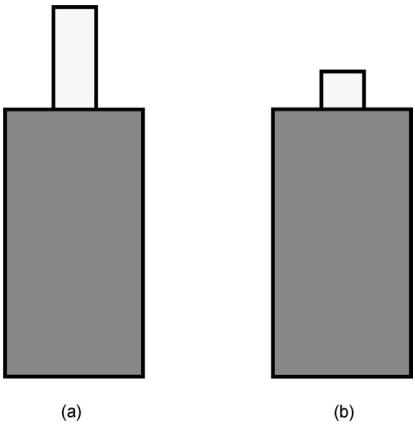


FIG. 6.46 Solenoid: (a) core normally out; (b) core pulled in by electromagnet

to the magnetic pull or push created (Fig. 6.46). Many solenoids have internal springs to return the core to the home position. They can buzz or hum when actuated.

They do, however, have a small range of motion, and they cannot handle any significant side loads. They do wear out after so many 10,000s of cycles, so they many not be sufficient for high-speed operation without significant preventative maintenance scheduling. It is not clear that a solenoid would have the strength and range of travel required for this application. Their motion limit is usually around 0.5 in.

Magnetic levitation may seem like a high-tech dream being applied to a down-to-earth problem. Maglev, as it is referred to for short, is used in Japan for high-speed trains. And would it require that the cases, now made solely from plastic, be modified to be magnetically actuated materials? Or at least a pallet or part of a conveyor is moved under the case each time a new case comes in to be unlatched?

But this perhaps silly discussion will help to raise a red flag not just for Maglev, but for the solenoid! The cases contain magnetically recorded audio tapes. Will Maglev, or even a simple solenoid, cause damage to the tapes? This concern is big enough to remove any use of magnetic manipulation from our considerations (Fig. 6.47). It even helps us ask the questions, "If we use an electric DC motor, where will it be placed, and will it be too close to the cases and cause tape degradation?"

So, adding what might be dismissed as a silly or even stupid suggestion to the list on possible workstations sometimes does bring out the red flags early on, when it does not cost any money to change a potential problem. Imagine if after

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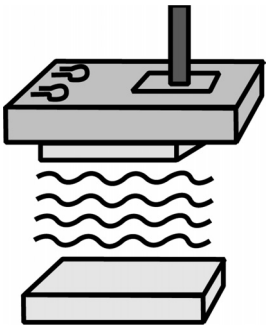


FIG. 6.47 Maglev lifting of the case: tapes would be damaged?

building the machine one finds that the DC motor is directly next to the case at one point in time, and by activating the motor one causes multiple gaps in the audio tapes. One might not even get word of such a problem for weeks or months after the machine is supposedly installed and paid for, and then this bug would tend to drive one crazy, and be very costly to debug and repair!

6.4.5. Using a Four-Bar Linkage

The use of a four-bar linkage would seem initially to be a poor choice. As seen previously in Fig. 4.19, the general motion of any coupler path is a cross between circular motion and a deformed banana. It does not normally lend itself to straight line motions. There are some special mechanisms known to designers for generations (Norton, 1999; Chironis, 1991) that do generate straight line motion by using only rotational joints, but these do not apply to this problem for several reasons:

- The required space for such a linkage is quite large, and makes little sense.
- The costs of the many rotational joints far surpass the costs of two linear bearing rods.
- Repair and maintenance would be more costly.

However, this design can work, and would be useful if this was part of a synchronous machine with a common drive shaft, in particular if we move away from a four-bar linkage toward a slider crank mechanism. This is how pistons in your car's engine work. These are also well documented (Norton, 1999; Chironis, 1991). With a slider crank mechanism, the input rotation of the crank is transformed into linear motion of the slider. The crank rotation would then need to be controlled to stop when the case had reached its upper limit sensing, and then most likely reversed so as to lower the case after latch unfastening. This reversing

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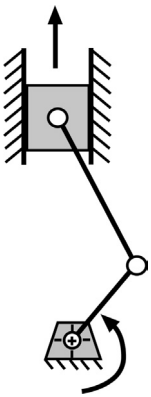


FIG. 6.48 A slider crank mechanism

might seem like more work than it is worth, but remember the DC motor also needs to reverse in the previous designs for linear lifting and the scissors jack.

A slider crank mechanism (Fig. 6.48) would produce the same effect as an air cylinder, yet the slider output most likely has already been designed using a single linear slide. This slide might be beefed up to have two slides or a double slide, so when the case is lifted, all side loading would be handled within its specified limits. A slider crank does have higher cost than the linear sliding device, since there are additional members and bearing supports (Fig. 6.49).

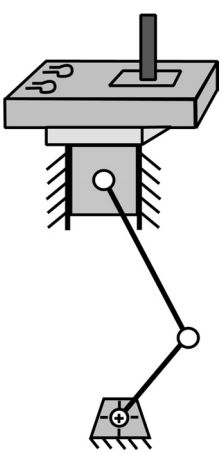


FIG. 6.49 Lifting using a slider crank mechanism

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6.4.6. Conclusions

The deck of cards stating possible workstation concepts was stacked in this case study to present the two better solutions first, but we have seen that the consideration of the other concepts, if even for only a few minutes, has provided some additional insight and red flags. There are still many details for this case study (as well as each specific project) such as the CAD layout of all parts and a complete bill of materials, which we will leave for Chapter 12 on Specifications.

6.5. CASE STUDY NUMBER 2: AUTOMATED SCREWDRIVER WORKSTATION DESIGN

Depending on one's environment, one can be exposed to screws and screwdrivers at an early age. Predominately, the world seems to be assembled with slotted head screws (using a flat edge screwdriver) and Philips head screws (that look like a plus " + " sign). Usually one finds a Philips head screw and screwdriver the easier to use since the screwdriver is less likely to slip off of the screw's head, but with either type of screw, a manually operated screwdriver is something that most people can use.

Now imagine you are to make a workstation to insert and tighten screws. From your own experiences, you think that this would be a relatively easy task. Why even a child can use a screwdriver, or so you think, but let us look at the steps required:

- Pick up a screw from a table top or from a package of multiple screws.
- Align the screw with its head on the top.
- Insert the screw into the previously drilled hole.
- Align the screwdriver blade to the screw head.
- Twist the screwdriver and apply a downward motion.
- Stop when the screw is snug, but do not overtighten.
- Remove the screwdriver from the screw head.

This list probably seems longer than what you would normally think about while you were performing these steps. Many of these steps you would take for granted. It is these "take for granted" steps that are usually the more difficult for automation to replace the human effort.

If this task was to be performed by a human operator for a significant part of one's work shift, then the person would probably use a powered screwdriver rather than the simple manual one (Fig. 6.50). The screwdriver could be powered by an internal electric motor and battery, or it could have an electric power cord attached so as to not be dependant on batteries, whether rechargeable or of standard type. The electric screwdriver could also be configured with an adjustable

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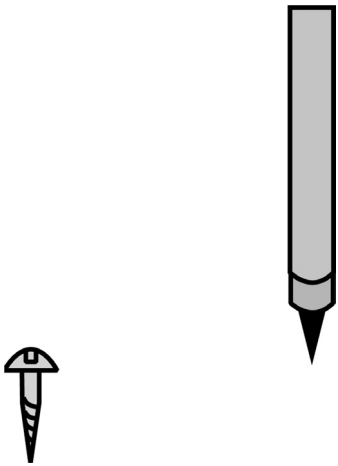


FIG. 6.50 Traditional slotted screw and powered screwdriver

slip release clutch to have a repeatable torque release when the screw has been completely inserted and is at the desired amount of snugness.

The screwdriver could also be powered by pneumatics (compressed air). These types of powered screwdrivers are not usually found in one's home toolbox, but are common in industry. Similar to the model using an electric power cord, there is no downtime as batteries get weak and replaced, and compressed air might be safer in the assembly of some items than electricity.

There are clips found on some home-type screwdrivers that will hold the screw head onto the blade while the screw is being twisted into the hole. The clips slip away from the head (most times) as the screw head comes in contact with the surface. The benefit of holding the screw firmly during insertion, particularly if the process is inside a complex environment, is worth the effort of sometimes having to pull the clip away from the screw head before giving the screw the final twist.

However, Fig. 6.51 shows the initial environment for automation. Many a student in an automation laboratory will solve the problem of obtaining a screw and orienting it correctly by placing a series of screws into a rack that has holes in it for just such a purpose. We have just created another task to be either humanly supported, or need another smaller automation workstation, that of filling the rack with screws.

The issues are now how to grip the screw head and align the slotted blade with the slotted head. One might think that the problem would be simpler with Philips head screws (and it would), but some applications wanting to automate

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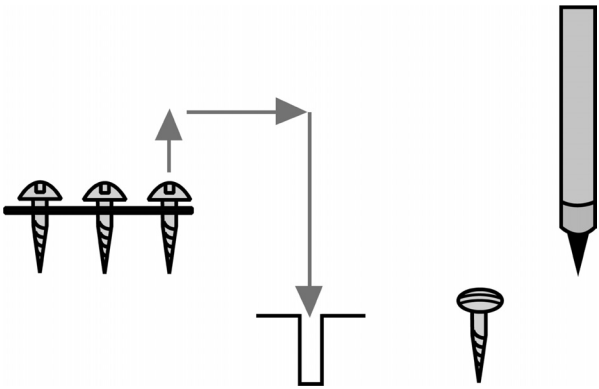


FIG. 6.51 Screw feeding issues and driver head alignment

still select fasteners that are not automation friendly. Grips similar to the hand held model can be used, and even the Philips head needs to get aligned to a degree. So some type of spring-loaded device is often used that allows for the blade to spin in contact with the head for so many revolutions until the blade and head must be aligned.

Students (and other researchers) have investigated magnetic forces to hold screws onto the screwdriver. But these types of forces are not very robust if the automation system slightly bumps the screw into the side of the hole while the insertion is being processed.

6.5.1. Automatic Screwdriver Workstations

Most commercially available powered screwdriver workstations have evolved using a different strategy than the human operation. They use a vibratory bowl feeder (see Chapter 7) to create a stream of screws, all pointing in the same direction (usually point first). The stream of screws is then conveyed to the workstation area using a hose and compressed air. Figure 6.52 shows the basic concept.

The screws are blown into place when needed. There might be a movable gate to keep the next screw in line from falling into place too early, which would prevent the screwdriver from properly engaging. A set of grips similar to the hand-held model stop the screw from being blown out of the system entirely. The screwdriver blade then travels forward, spinning to align the blade and the screw head. The motion device (robot or other transfer arm) carrying this automatic screw machine would then move into location above the desired hole. The screw would be rotated until the screwdriver clutch would slip. The

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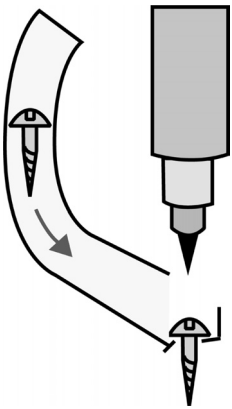


FIG. 6.52 Automatic screw feeding and screwdriver clutch

grips can be retracted if needed to avoid being stuck under the screw head. This means the correct amount of torque has been applied. Then the motion device would move to the next hole while the next in line screw advances into place.

6.5.2. Conclusions

The workstation in this case study has two major components. First is the bowl feeder to supply the screws in the correct orientation. The second component is the screwdriver head that fits at the moving end of a robot or transfer arm. This case study is significantly different from the earlier one, since there are two components only joined by the air hose carrying the screws, and that it is a situation where one should not design their own workstation.

6.6. MACHINE DESIGN AND SAFETY

You are responsible in the real world for the safe operation of any machine that you design and/or build. Whether you know about applicable safety standards or not, if there is an accident involving your machine, the courts may find you liable. Ignorance is no excuse. Machines can be designed with safety in mind, or they can be corrected later with external guarding, and so on, but this can be a life and death issue that can haunt you for decades to come. It has put companies out of business.

Early machines, like some of the lumber mills powered by waterwheels 200 years ago, are a great example of how NOT to do safety. Overhead common drive shafts were coupled to giant saw blades by wide leather belts. The belts had no

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cover or shield to prevent someone's clothing from getting caught between a pulley and a belt, so this could be a gruesome situation without even coming into contact with the saw blade. And then the saw blade had little or no guarding itself, in comparison to today's radial arm saws. Accidents were often seen as a way of life. Lawsuits did not rule the day as in the present. There were no advertisements on television for lawyers who promised you big bucks when you had an accident at the lumber mill. Power takeoffs on farm equipment are notorious for fatal and mangling accidents, too.

Now everyone in the chain of machine designer, builder, installation group, maintenance team, and so on can be a party to a lawsuit. It is unfortunate but unavoidable today. One needs to do homework to have a strong case so as to be able to walk away without a judgment against one. This starts with knowing what the safety standards are, and also applying common sense.

The concept of designing a safe machine can be found as early as 1914 in the Universal Safety Standards (Hansen, 1914), where the Workman's Compensation Service Bureau stated for steam engines that "All dangerously located moving parts, such as fly-wheels, cranks, eccentrics, cross heads, tail rods, fly balls of governor, governor sheaves, etc. to be guarded in approved manner." Now some of these devices, such as mechanical governors, are no longer used in today's machines, but a similar statement was made for power transmission (Hansen, 1914). So the message has been clear for decades. If it moves, it should be guarded.

An early textbook by Hyland and Kommers (1929) states that high-speed machines are a hazard, and need to be guarded. They document that in 1927 the American Engineering Standards Committee adopted a safety code for Mechanical Power-Transmission Apparatus. Their guarding statement was augmented from the 1914 version stating that guarding needs to go all the way to the floor.

6.6.1. Pinch Points

The classic problem for safe machine operation is defined by the term "Pinch Point." Figure 6.53 shows three different occurrences. The location where two rollers press something, called the nip, is a standard situation. This is where any loose clothing can get drawn into the rolling and drag the wearer into it too. The moving lever also can come down and do some harm. Often there is some gearing or other mechanical advantage being applied, so a human cannot arm wrestle the lever with much success. The third situation is often an oscillating press. Again, due to the processing requirements, the forces involved are life threatening.

Guarding these problems is a must. None of us can claim to be born and practicing engineering before 1914, so one cannot claim guarding is only a new fad no-one knows about. And to add more evidence that the standards for

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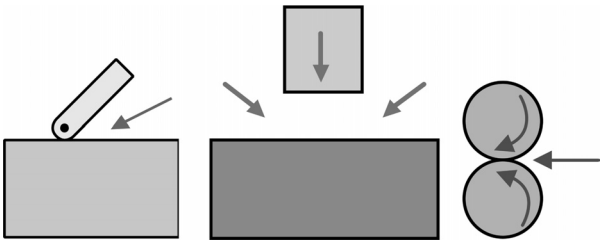


FIG. 6.53 Pinch points

safety do not stay fixed on 1914 definitions, the American Society of Mechanical Engineers (ASME) and American National Standards Institute (ANSI) have established and revised safety standards for Mechanical Power Transmission Apparatus (MPTA) (ASME, 1994). Because of the fact that most if not all automation machines have power transmission somewhere within the machine framework, the MPTA standards apply.

To make things even more clear for some automation machines, there are ANSI standards for Packaging and Packaging Related Converting Machinery (ANSI, 1994). Current versions can be inspected on their website (www.ansi.org). There are safety codes for robots (ANSI/RIA, 2003). There is a great deal to be concerned with depending on the specific machine and application, but guarding or shielding is a basic must.

The timeframe of when the machine was first designed and built is the most critical. A machine designer who incorporated all of the current standards of the day will most likely not be faulted if an accident occurs that deals with more rigorous standards 20 or 30 years later, but if the 30-year-old machine is still being produced for all of the 30 years, and more rigid safety standards are adopted, the issue of whether the manufacturer needed to issue a recall, or simply a warning advisory to all known customers, is a situation one should involve a lawyer to decide. In a court of law, only the standards of the day can be included for consideration of whether it was a good or negligent design.

Guarding can be in the form of a permanent wall, but access is often required to clean up leaking grease or clear out a jam. Figure 6.54 shows three ways to handle this situation. A door can open to one side. A cover can lift above, usually assisted by springs, sealed air cylinders or counter weights. Or a series of light beams can be used to create a curtain that will not allow a human hand to enter without shutting down the machine's operation. The physical moving doors usually incorporate a sensor interlock switch and control system.

There is always the need to get into these critical moving parts, if one asks a machine operator or maintenance person, and although it is against all safety

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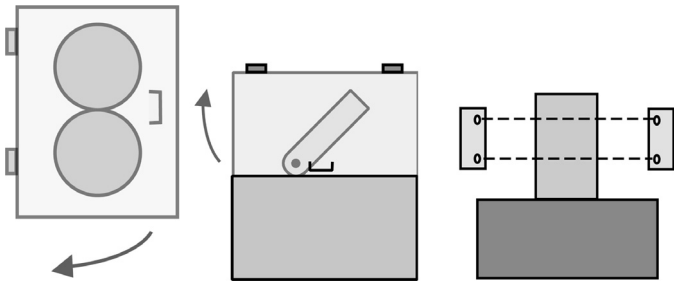


FIG. 6.54 Shielding pinch points and moving devices

standards, some operators get tired of proper shut-down procedures, particularly if the machine jams every 15 minutes. So sometimes safety interlocks are defeated or bypassed, and the consequences are troubling. Never allow for any safety bypass situation without proper backup procedures.

6.6.2. Lockout/Tagout

Before the controller's front cover of an operating machine be opened, one should investigate if the type of wiring and power usage requires a Lockout/Tagout arrangement. This means that the energy sources, usually AC electric current, must be turned off before the door can be opened. The lever that switched the power off may be coupled to a physical door locking device. Lockout/Tagout (ANSI, 1982) lists how and when the lever gets padlocked, and the labeling required to let everyone know what has been turned off, why it was turned off, and who has the key. A licensed electrician should know this process and assist you.

6.6.3. Warning Labels

It is very good to use warning labels (Fig. 6.55), but they are not sufficient. You still must guard a dangerous situation. However, warning can help. There are warning labels all around us now, so much so that we may be getting complacent about them. There are standards for the proper label type and location from ANSI and the National Electrical Manufacturers Association (NEMA) (1991).

6.6.4. Risk Assessment

The Packaging Machinery Manufacturers Institute (PMMI) in conjunction with the ANSI 2000 version of the B155.1 safety standards has embraced the concept of risk assessment (PMMI, 2000). Any safety standard sets the minimum level of

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FIG. 6.55 Warning labels

safety required. Machine designers and manufacturers need to do even more. They need to look at:

- What different types of machine occurrences can happen.
- What is the anticipated misuse of this machine.
- What is the level of possible injury.
- What is the probability of occurrence.
- What can be done to rectify the problem.
- Document the process and results.

If this process is done with due diligence, it should greatly minimize the number and magnitude of safety-related issues. Done well, it will also speak well of the machine designer's efforts if and when an accident comes to court. Accidents are sometimes assigned by juries a certain percentage due to faulty machine safety design, careless people near the machine, and/or a freak occurrence of nature. The determination in the courts today may not always be logical, but doing one's homework will be better than not!

6.6.5. Safety Responsibility After Delivery

Very often, after one delivers an automatic machine to a customer, there will be several if not many opportunities to review the machine in operation over time. Then is when the designer, builder, and even the salesperson need to keep a sharp eye towards how the customer is using the device. One should inspect to see if the customer has removed any guards, or defeated any safety interlocks. If so, the customer should be notified by some type of certified mail that this practice is not endorsed by your company. Mr. Fred Hayes summed this up well in two recent PMMI documents (Hayes, 2003; PMMI, 2003), when he stated "If you see it and condone it, you own it."

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6.6.6. Safety Standards are Just a Starting Point

National safety standards are viewed by many to be a bare minimum. Many large companies have more stringent requirements than OSHA or ANSI. One needs to be aware of these before quoting a system. Other times in court cases for injury lawsuits, experts are questioned if the applicable safety standards are sufficient. If one explores the situation for any length of time, one may conclude that there are other seemingly common sense items that need to be guarded or accounted for. Energy can be stored in a spring or compressed air and can do damage after the machine is turned off. Do not simply rely on safety standards. Look at the risks involved! Is it obvious that someone might try to defeat a safety interlock because of the need to adjust the equipment while it is running?

6.6.7. Real-Life Accidents

The author has inspected more than 20 potential machine safety failures related to product liability lawsuits, both for the injured party and the machine builder. Some of these accidents seemed likely to occur someday, such as when an operator is daily to clean grease from a moving chain and sprocket near a series of knitting needles. Other accidents seem highly less predicable, not that the root cause was hard to identify, but the resulting accident (as witnessed by several people) was from a chain reaction similar to one only seen on Saturday morning cartoons. Not that the accident was in any way funny. However, the chance happening of one unguarded device hitting a second obstacle that in turn moves and injures someone 30 ft away, would seem impossible to anticipate or reproduce.

Safety needs to be incorporated into the design of automation in today's society. One needs to think of all possible ways for misuse or stupidity. To wait until it is all constructed, and then to guard it, knowing there will be operators reloading or cleaning it, is just foolhardy.

6.7. CONCLUSIONS

Workstations are when the concepts and dreams become reality. This process of detailed work is generally not as creative or as much fun as the earlier stages, and some student teams lose their excitement. However, without good thinking and investigating, you will find yourself using the equivalent of Maglev and within week have thousands of damaged tapes. The real engineering heroes earn their rewards at this point.

PROBLEMS

1. Create a workstation to mix the chocolate chips into a cookie batter for your home use. Determine what would happen if one used their home lower powered hand mixer, and the batter was fairly thick. Determine

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the impact temperature has on the batter's viscosity. Would the need for this workstation be the same in both the Arctic and at the equator?
2. Investigate on the Web for available workstations to perform various tasks required by students or engineering employees.
3. Design a workstation to automatically butter your toast as it leaves your toaster. Determine the relative needs for:

    - structural members;
    - bearing surfaces;
    - drive mechanisms;
    - types and ranges of motions;
    - general sensing needs;
    - safety.

PROJECT ASSIGNMENT

Using one of the projects from the Appendix (or any other projects), perform the following:

1. Create some concept sketches for the needed workstations. List the requirements for the:

    - structural members;
    - bearing surfaces;
    - types and ranges of motions;
    - general sensing needs.

2. Compare the concepts and lists from step 1 above and perform a matrix comparison, stating the pros and cons for each concept. Create a list of the unknowns or the areas of additional needed research or investigation. Determine what parts of the process seem to have some risks associated with it.
3. With the associated risks and benefits of the workstation concepts in the steps above, determine the most likely set of workstations and machine configuration. If this is solely an academic exercise, you may wish to go ahead with a choice that has more risk but will be a richer learning experience. In real life, you may want to limit your exposure to risk.
4. Make an initial safety review of your leading design. Determine where and how shielding should be implemented. Walk through the normal operation process, and any re-supply or error correcting actions and list possible concerns. Try to predict when, how and why someone

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might want to defeat your safety system, and see how you can anticipate and account for this.

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